

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

November 2012

CL211 Environmental Engineering-I

Date: 08.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 (a) Discuss salient points considered for site selection of a water treatment plant? [05]

(b) Define following term (any one). [02]

(1) pH, (2) displacement efficiency, (3) detention period

Q-2 (a) Enlist various methods used for population forecast and explain geometric increase method in detail. [05]

(b) The following is the population data of a city. Determine the population of city in 2011 by arithmetic increase method. [05]

Year	1931	1941	1951	1961	1971	1981	1991
Population (P)	13000	18500	28500	42600	60100	69800	75200

(c) Write short note (any one): [04]

(1) Flash mixer (2) Design period (3) Flocculation

OR

Q-2 (a) What is coagulation? Enlist various properties of a good coagulant. [05]

(b) A water treatment plant treats 300 m³/h of water. Find dimensions of and power required for a rapid mix unit. Assume suitable values of detention time and velocity gradient G. Take dynamic viscosity $\mu = 1.0087 \times 10^{-3}$ Pa.S [04]

(c) Determine the quantity of alum required to treat 15 MLD of water if dose of alum required is 12 mg/L. If chemical formula of alum is $Al_2(SO_4)_3.18H_2O$ and it reacts with bicarbonate alkalinity ($Ca(HCO_3)_2$) in water, calculate the mass of carbon dioxide that will be released per liter of water. Molecular weights are: Al=27, S=32, O=16, H=1, Ca=40, C=12. [05]

Q-3 (a) For a horizontal flow rectangular sedimentation basin, prove that settling velocity of a particle is independent of basin depth. Also write the assumptions made. [05]

(b) Find out settling velocity of a silica particle of diameter 0.004 cm and specific gravity 2.67 in water. Take ν (nu) = 0.009 cm²/sec. [04]

(c) A primary settling tank is 25 m in diameter with side water depth of 2.1 m. Single effluent weirs are located at the periphery of the tank. If flow of water is 26000 m³/d, calculate: 1) surface area and volume, 2) surface overflow rate, 3) weir overflow rate, and 4) detention period. [05]

OR

Q-3 (a) Discuss briefly various inlet and outlet arrangements made in a rectangular sedimentation tank. [06]

(b) Design a rectangular sedimentation tank to treat 3 million liters of water per day of [05]

coagulated water. Assume detention period to be 4h and horizontal flow velocity = 100 mm/min. Assume suitable water depth. Also calculate the surface overflow rate and weir loading rate for this tank.

- (c) Enlist various sources of water and impurities normally present in each. [03]

SECTION - II

- Q-4 (a) Discuss various factors to be considered for site selection of an intake structure. [05]

- (b) Draw classification chart for intake structures. [02]

- Q-5 (a) With neat sketch explain working of a bell mouth canal intake. [04]

- (b) What is hardness? What are its effects on household and industrial use of water? [04]

- (c) Explain with neat sketch working of a rapid sand filter [06]

OR

- Q-5 (a) Describe various actions that take place when water is filtered through a bed. [06]

- (b) A city has a population of 120000 with an average rate of demand of 140 LPCD. Find the area of rapid sand filter required. Assume suitable rate of filtration. If area of a bed is not to exceed 40 m², calculate number of beds required. [04]

- (c) Write short note (any one): (1) Effective size and uniformity coefficient, (2) backwashing in a rapid sand filter, (3) disinfection [04]

- Q-6 (a) Name various disinfecting agents and discuss ozonation in detail [05]

- (b) What are various systems of plumbing for drainage in buildings? Describe one pipe system in detail. [04]

- (c) Enlist various water distribution system layouts. Describe any one with neat sketch. [05]

OR

- Q-6 (a) What is ventilation? What is the purpose of ventilation? [03]

- (b) A city is to be supplied with 50000 m³/d of water. The difference of elevation between the city and water treatment plant is 40 m. If the total volume of water is to be supplied in 20 hours, calculate the diameter of pipe. Calculate BHP of pump if total pipe length is 2000 m. Assume velocity in pipe $v = 1.5$ m/s, co-efficient of pipe friction, $f = 0.03$, and efficiency of pump $\eta = 85\%$. [06]

- (c) Write short note (any one): (1) house drainage system, (2) types of traps, (3) air conditioning [05]

List of Useful formulae

$Pn = P + nl$	$P = \mu G^2 V$	$P = \frac{1}{2} C_d A_p \rho \cdot v_r^3$	$V_s = \frac{gd^2}{18\nu} (Ss - 1)$
$DT = \frac{V}{Q} = \frac{LBH}{Q}$	Pipe flow, $Q = \frac{\pi}{4} d^2 \cdot v$	$H_f = \frac{flv^2}{2gd}$	$HP = \frac{\rho QH}{75\eta}$
